



MINING DEALER BEST PRACTICE

Front Spindle Salvage on Vertical Lathe

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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June 2023
4.03-230607-1

1.0 Introduction

This Best Practice was developed with the purpose of making an optimal, effective, and reliable salvage in the front wheel spindle of trucks, guaranteeing its specification according to the criteria CAT Inspection and Salvage of Front Wheel Spindles for Certain Off-Highway Trucks (SEBF9309-10).

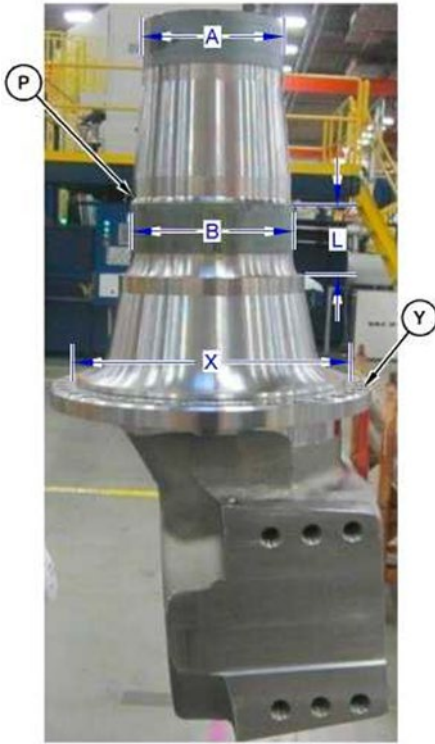


Illustration 39 g06452141
Spindle / front axle dimension locations, refer to Table 10 for the dimensions.
(A) Bearing Journal
(B) Bearing Journal
(L) Bearing Journal Length
(P) Chamfer
(Y) Face

If bearing journals (A) and (B) are worn, thermal spray may be used to restore the diameter of the bearing journal to the original dimension. If you have questions concerning equipment, recommendations for powder, machine settings, or procedures for the application, Refer to "Thermal Spray Procedures for OHT Front Spindles" section. Each step in the procedure is critical to achieve the desired bond coating and the desired surface texture. Preparation of the spindle / front axle is critical to achieve the desired bond and the desired surface texture of the coating. After the spindle / front axle has been sprayed, mount the spindle / front axle assembly on a lathe. Zero the part on diameter (X) and face (Y). Machine the spindle / front axle to the dimensions that are shown in Table 10. Bearing journals (A) and (B) must be within 0.038 mm (0.0015 inch). Both bearing journal (A) and (B) diameters must be within the maximum Total Indicator Reading (TIR) of each other and face (Y). Refer to Illustration 39 for the dimension locations.

Identification of Areas that Wear

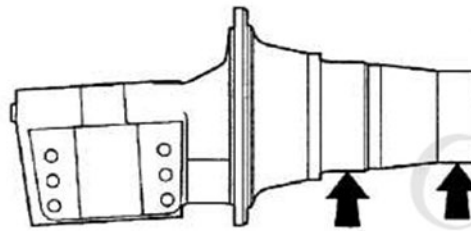


Illustration 8 g03694136
The area between the bearing and the spindle / front axle (arrows) may encounter wear.

Illustration # 01 and 02, The critical areas of the bearing's journals observed when they present wear spec out measure, they must be rebuilt with arc spray processes according (SEBF9309- 10).

2.0 Best Practice Description

In the machinery workshop of Barloworld Mongolia, we developed a device which allows vertical clamping of different front spindles Off-Highway trucks models 773 to 797, achieving an easy, efficient, and reliable salvage process in vertical lathes.

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3.0 Implementation Steps

3.1 Manufacture stand for mounting spindle front in vertical lathe, according to the image and drawings developed by the machine shop Barloworld Mongolia, they will be able to make this device that allows machining in vertical lathe in a more efficient, and reliable way.

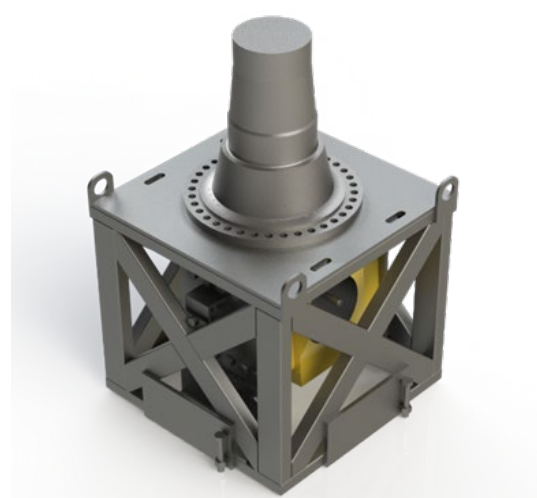
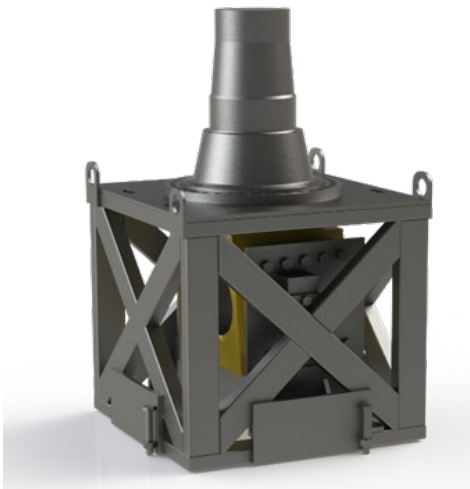
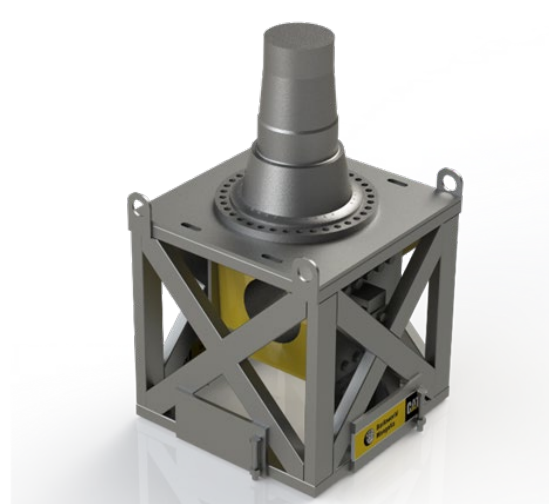
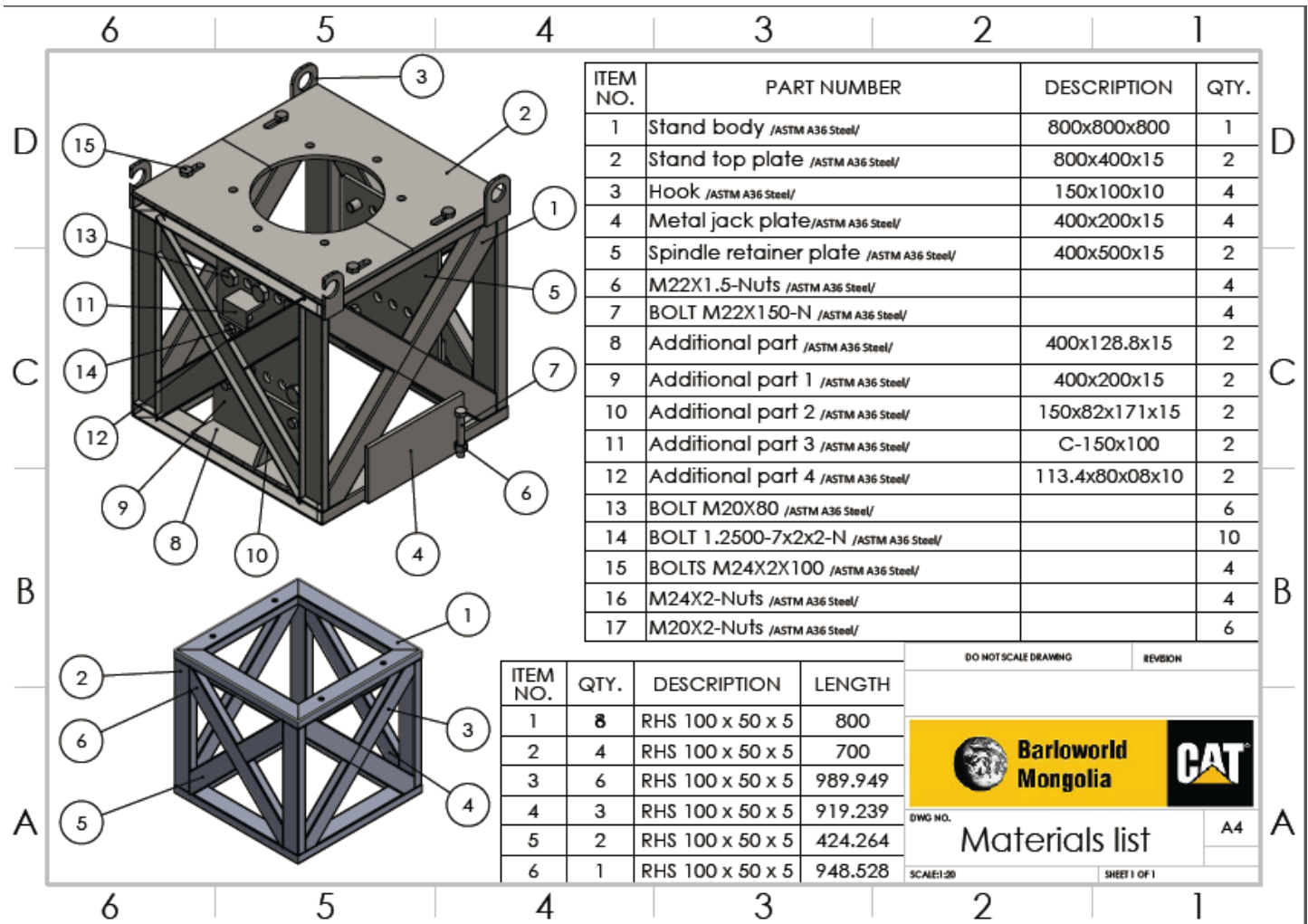


Illustration # 01,02,03 and 04 fabrication device according to the blueprint for positioning the spindle vertically.

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Material's list



The list of materials must be taken for manufacturing, the type of material is ASTM A-36.

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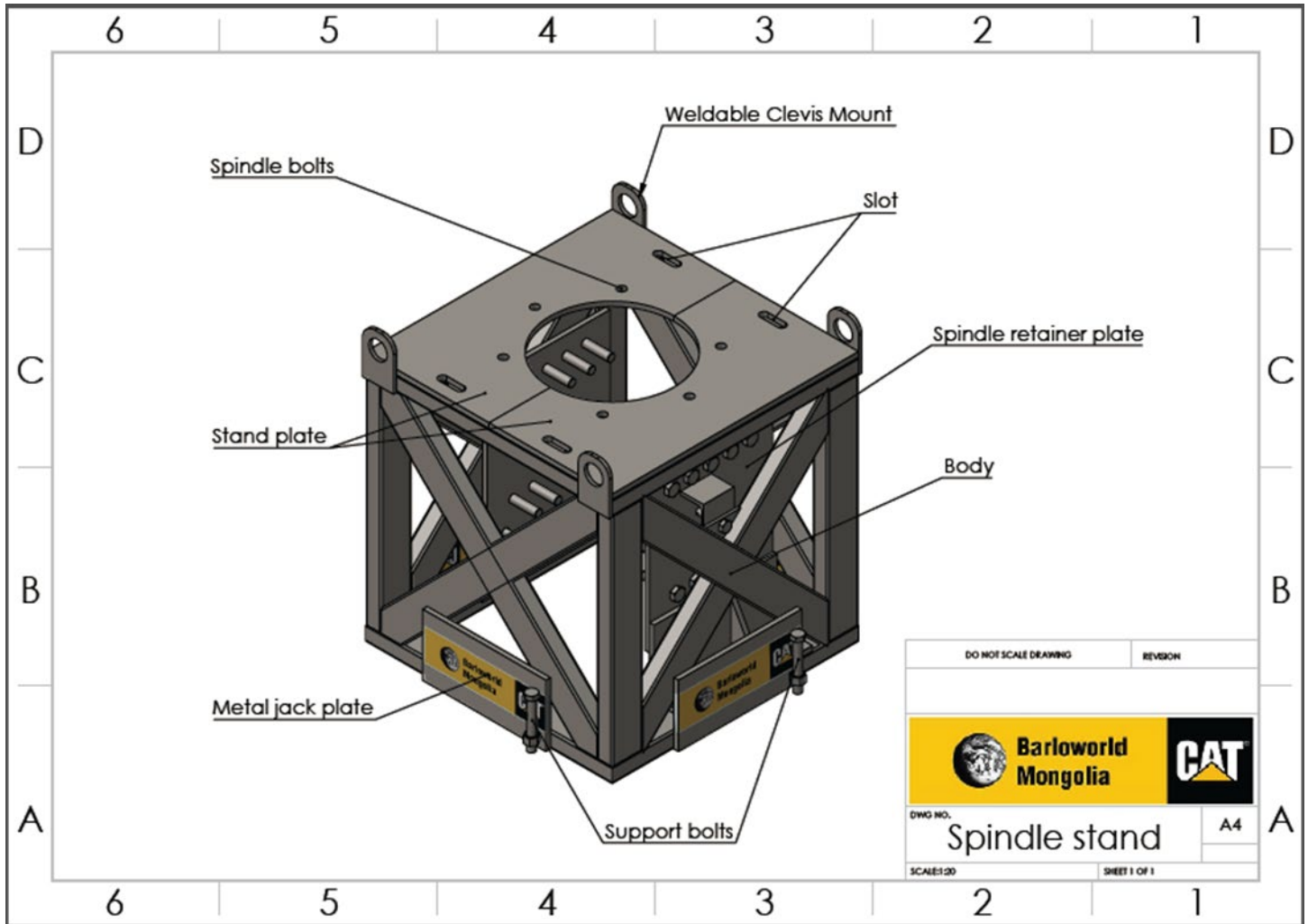
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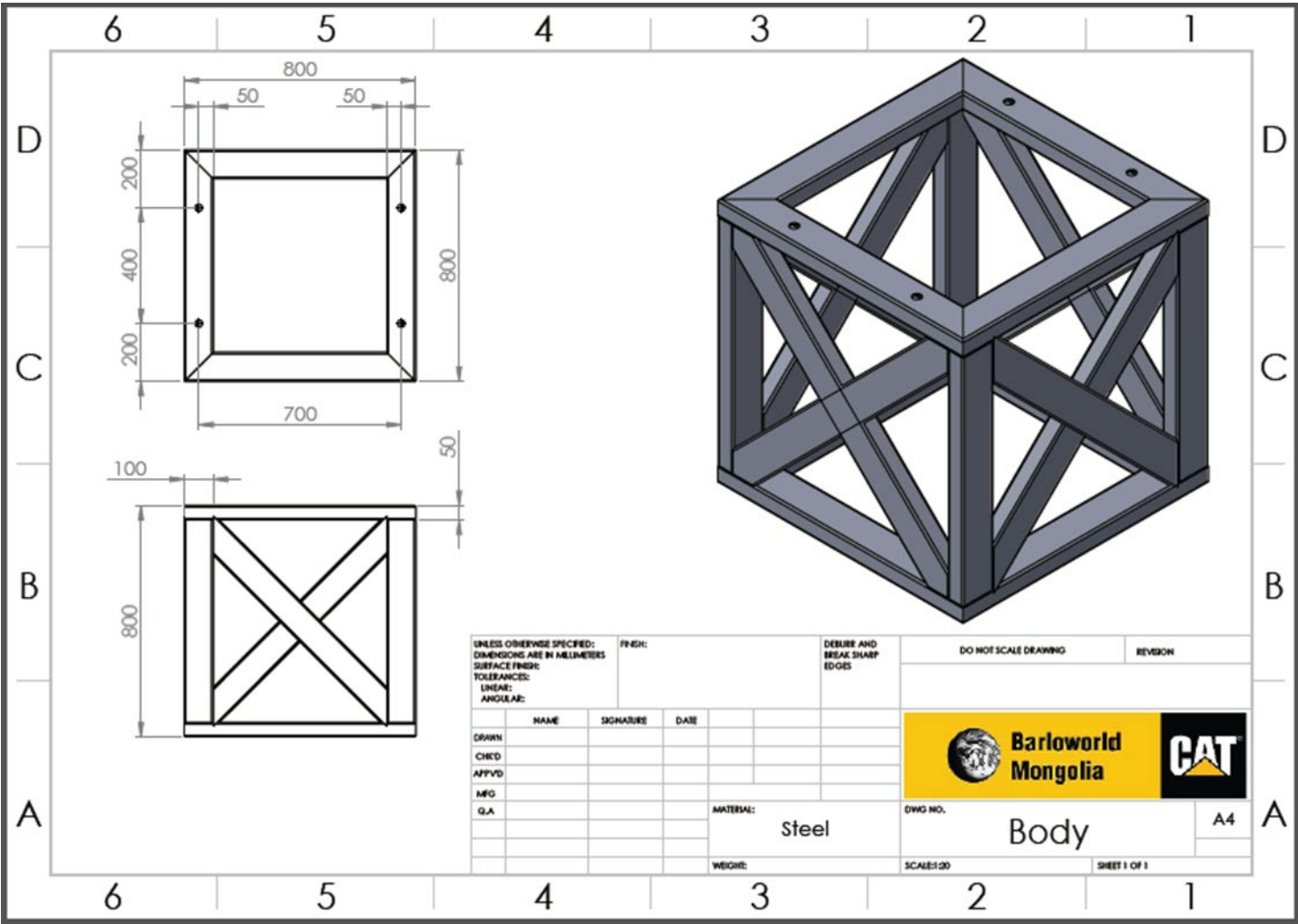
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Support Blueprint for Spindle Front Models CAT 785, 789 and 793



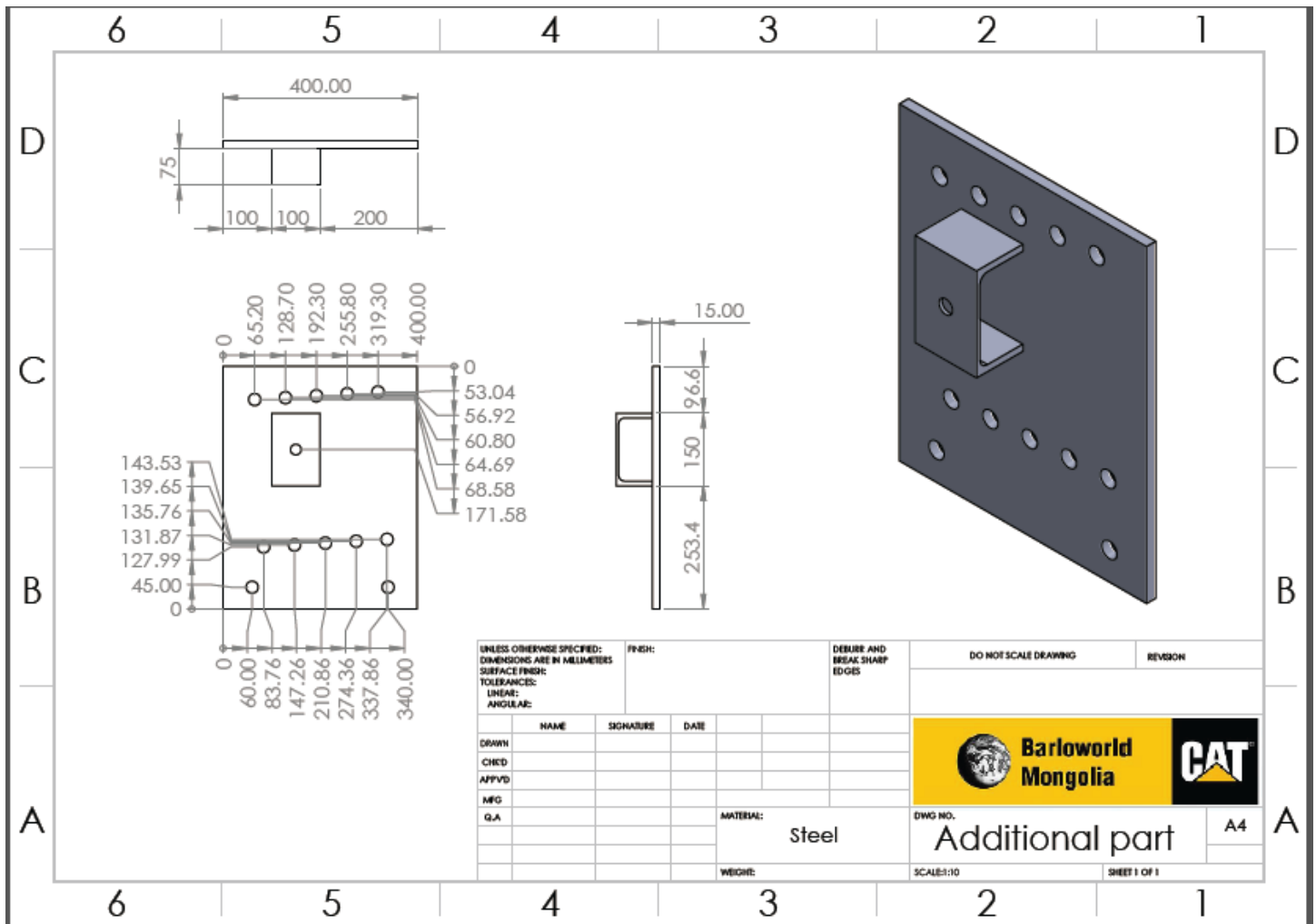
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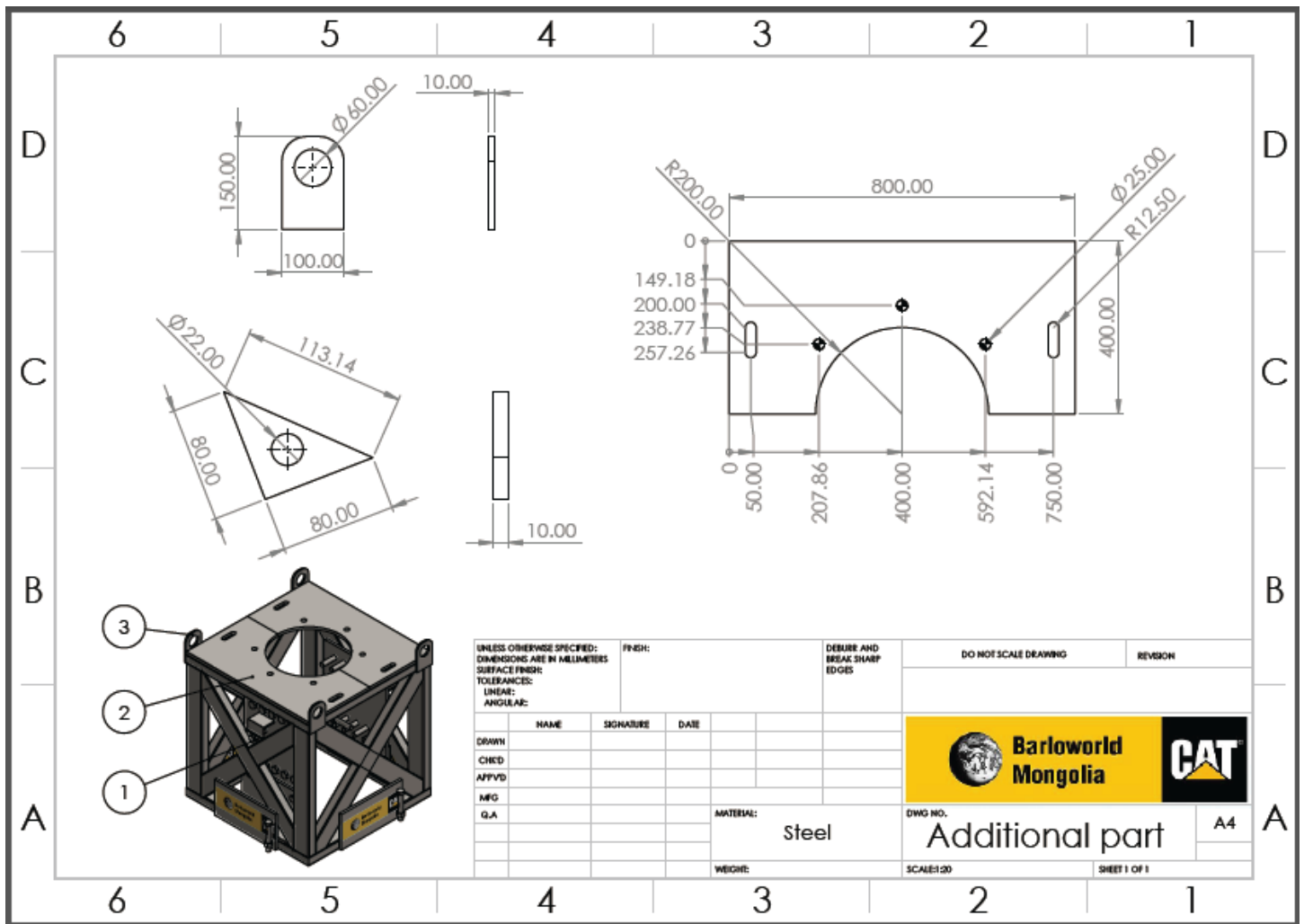
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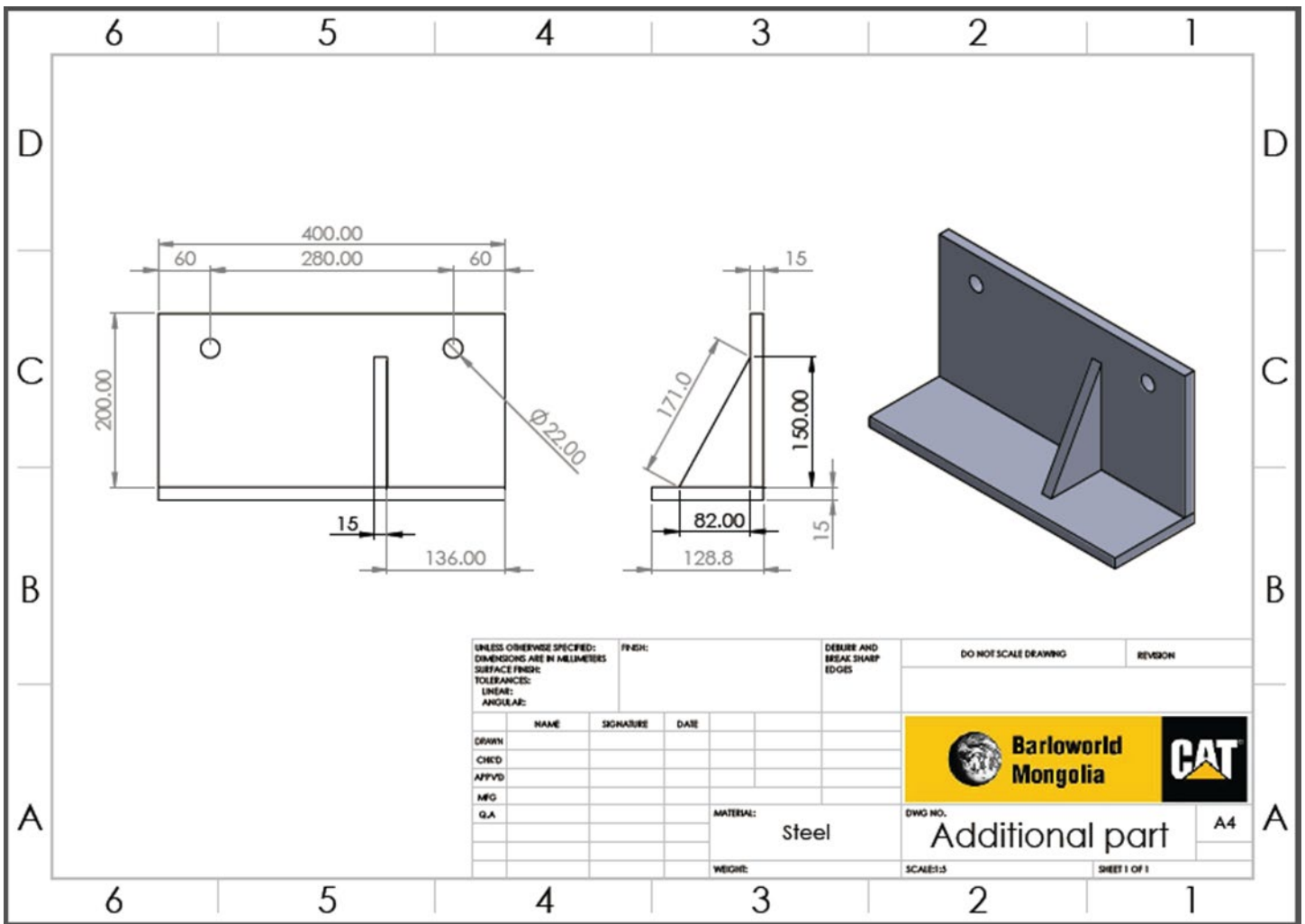
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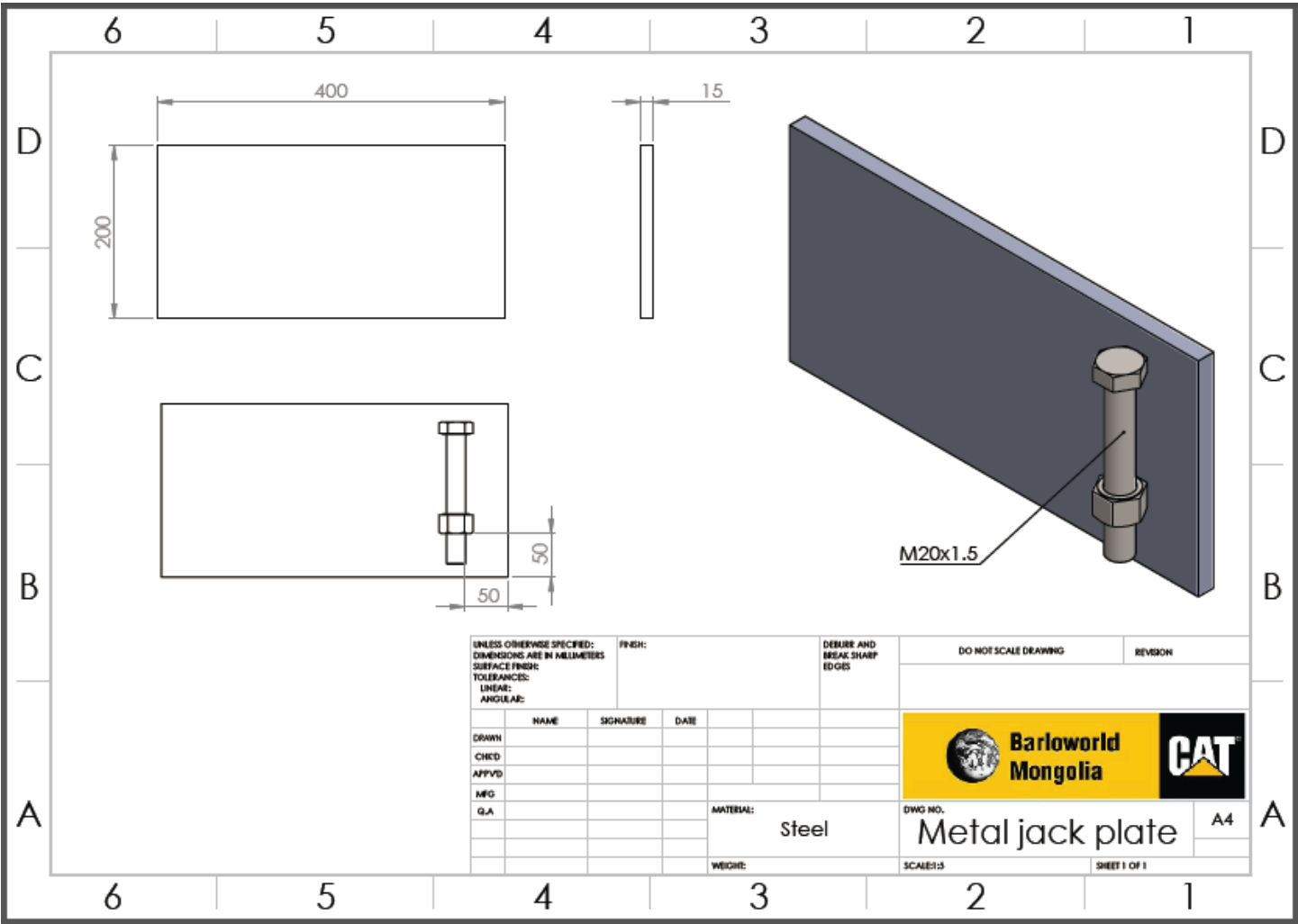
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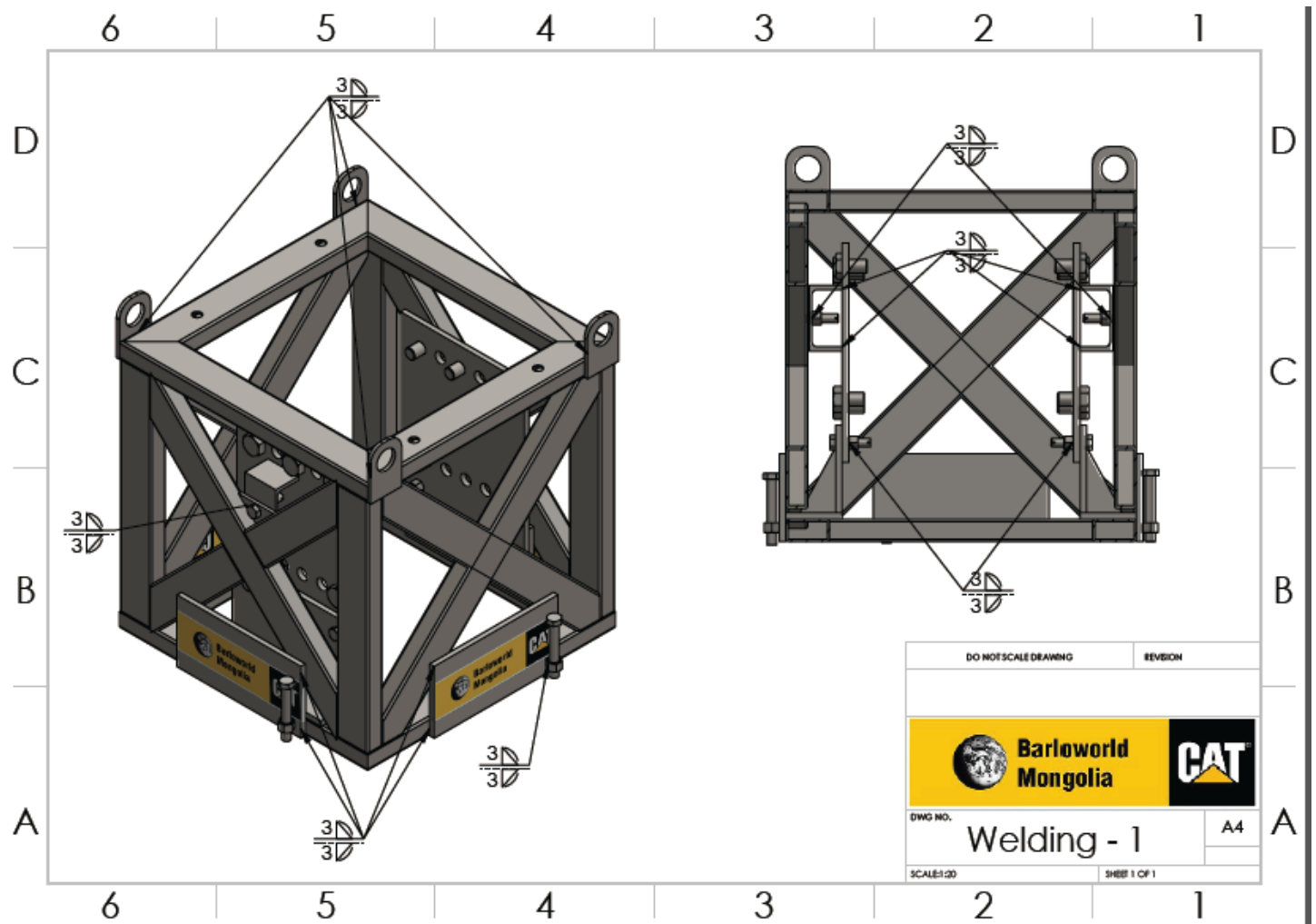
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Weld Details



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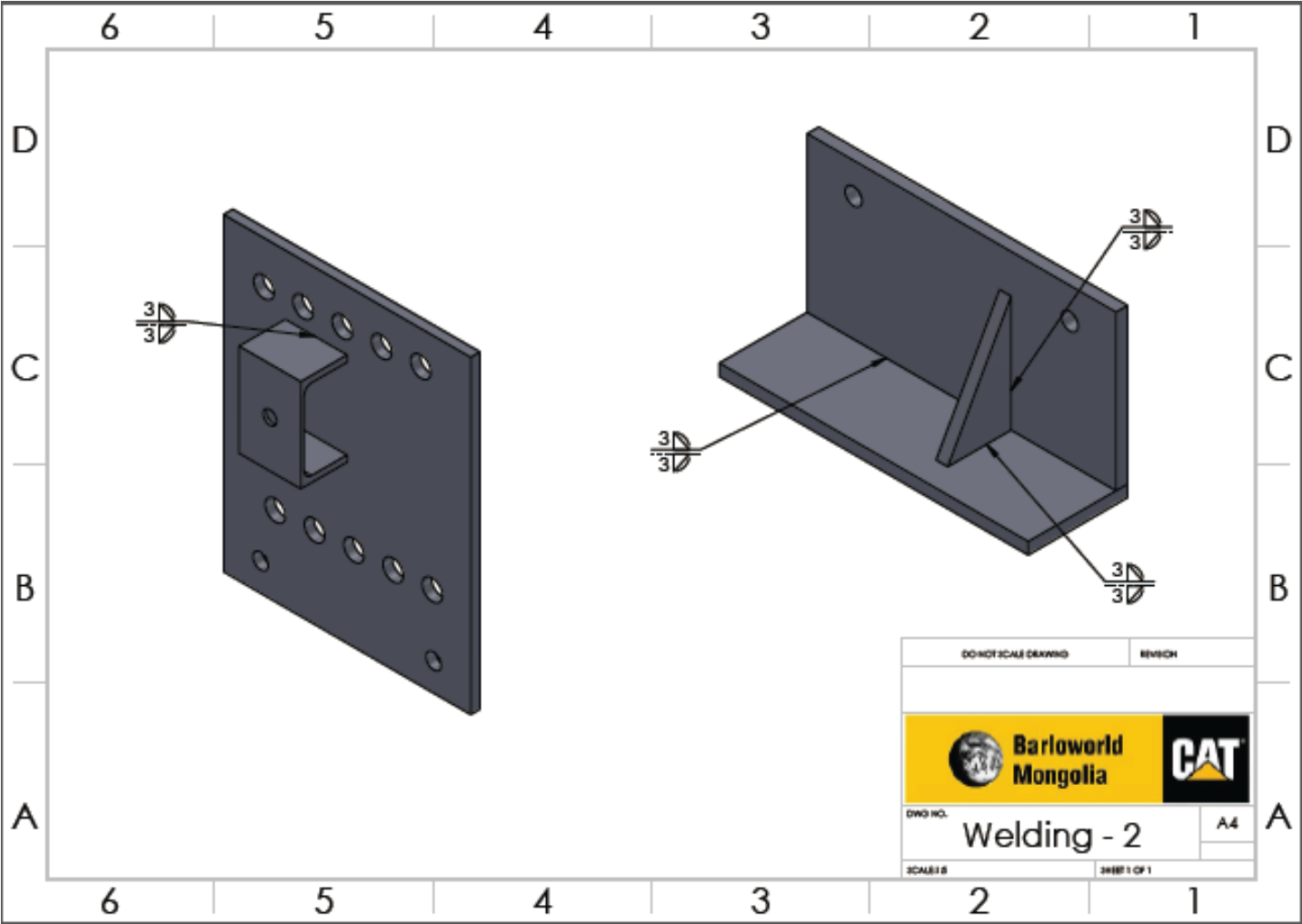
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Weld Details



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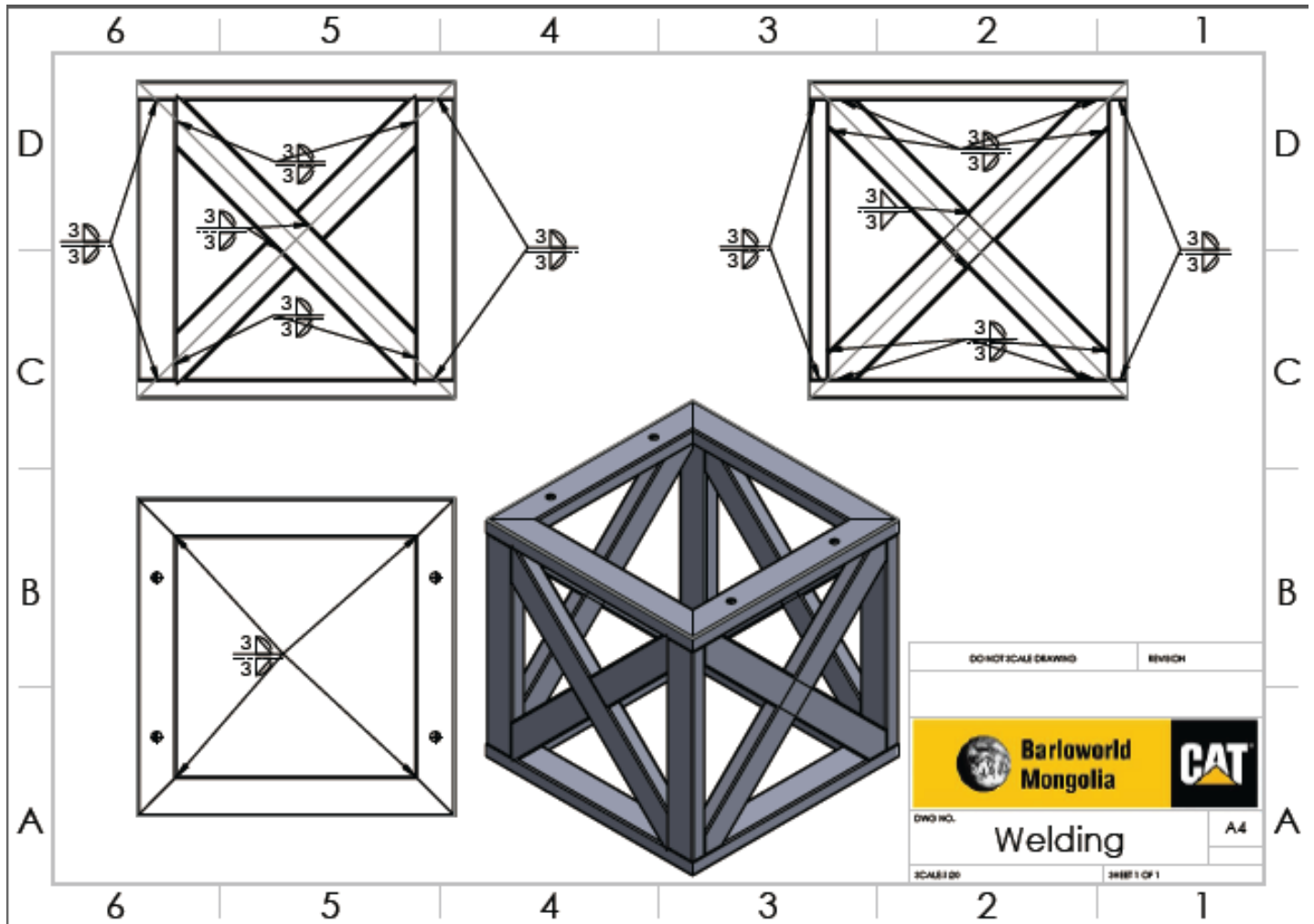
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Weld Details



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Stand material: ASTM A36 Steel

Total weight of Stand 405,9Kg or 894.86Lb



3.2 Pre-machining of the journal bearings -1mm of the nominal diameter with a rapid advance that obtains high roughness for the arc spray.

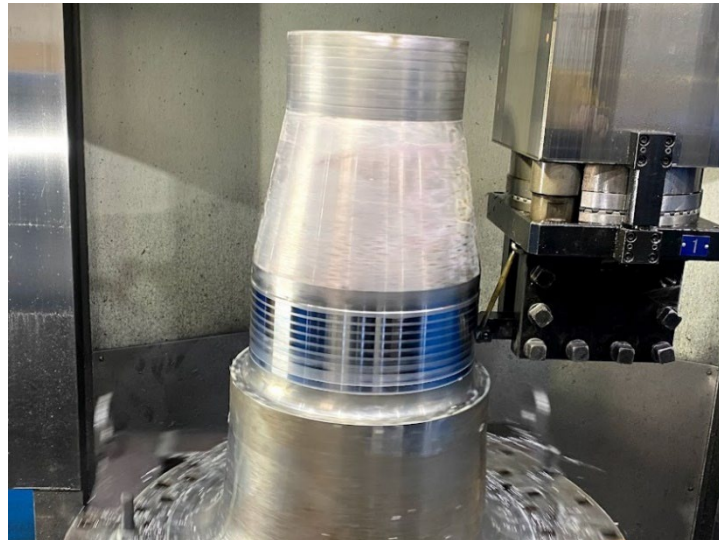
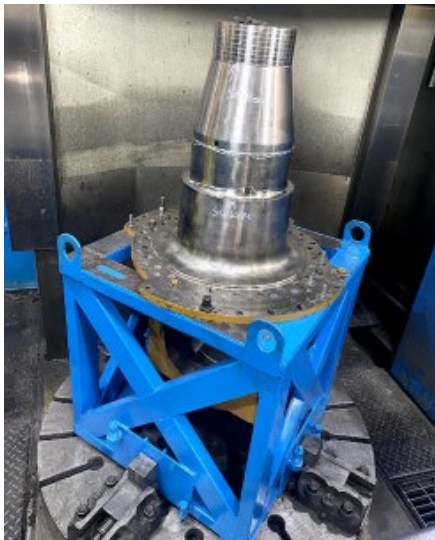
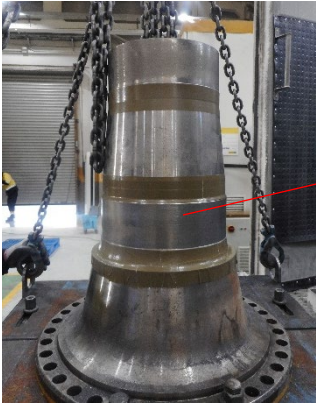


Illustration # 05 and 06 Perform 1mm diameter pre-machining on both journals bearing.

VARIABLE	OPERATION RANGE
Velocity	50rpm
Flat	0.35 mm
Cutting Depth	0.5mm
Cutting Tool	MDJNR4040P15
Program Name	cylindrical machining
Insert	DNMG150612MP KCU10

3.3 Grit Blasting After pre-machining, all the area must be protected, to apply Grit blasting in the pre-machine area, a roughness greater than 250μinch must be achieved.

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GRIT BLASTING PARAMETERS

Pressure	100 PSI
Distance	7-8"
Material	Metcolite C 400

3.4 Apply Arc Spray, SM 8400 or Tafa 75B wire base at a distance of 5" with a rotation speed of SMPM 40.00, the arc must be projected directly into the area to be recovered with a 45° angle then apply the coating, SM 8222 or Tafa 90MX wire at a distance of 4 "with a rotation speed SMPM 40.00, the arc must be projected directly into the area to be recovered with a 45° angle.



VARIABLE	OPERATION RANGE	
Wire	75b or 8400	8222 or 90mx
Voltage	30V	32V
Current	130Amp	140Amp
Pressure	60 PSI	60PSI
Distance between gun and workpiece	5"	4"
Spray Rate/Bond Pass	0.038mm- (0.0015")	0.038mm- (0.0015")
Traverse Rate of Gun	11.0 SMPM (40.00 SFPM)	11.0 SMPM (40.00 SFPM)
Rotation Speed of Part	92.0 SMPM (300.00 SFPM)	92.0 SMPM (300.00 SFPM)
Temperature	100-130°C	100-130°C

Illustration #09 Apply a coat of SM8400 or 75b Tafa base wire, then apply SM8222 or 90mx to ensure journal bearing hardness.

3.5 5 Final Machining, is done on vertical lathe machine or a similar one using the cylindrical machining program, cutting speed 50rpm advance 0.15mm int VNMG160404FF KCU10 Kennametal To carry out the final machining, it must be anchored to the work plate, center, and flange.

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VARIABLE	OPERATION RANGE
Velocity	45rpm
Flat	0.15 mm
Cutting Depth	0.5mm
Cutting Tool	MVJNR-4040 NG8
Program Name	Cylindrical Machining
Insert	VNMG160404FF KCU10

Illustration # 10 and 11 the final machining process under the CAT standard is observed.

3.5 Final delivery process, you must remove all the overspray, polish, clean and double check the measures for delivery.

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Illustration #12 and 13, Spindle delivery is observed.



Initial Evaluation



Arc Spray Process



Machining Finish



Polish Finish

4.0 Benefits

- 4.1 Improve component handling
- 4.2 Avoid ovaling in the machine
- 4.3 Greater efficiency in the process

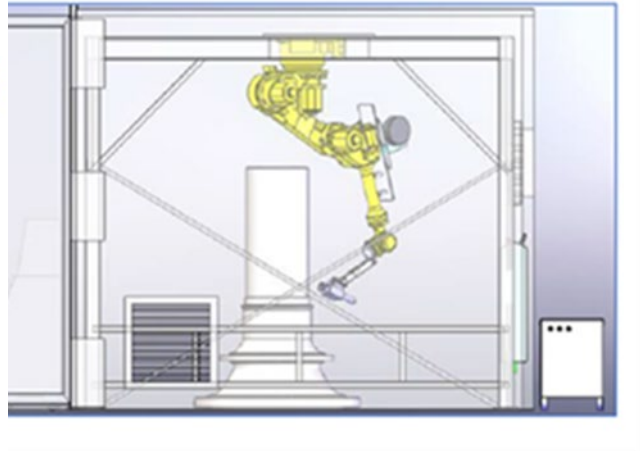
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5.0 Resources Required

5.1 Fully integrated, self-contained coating facility with a Smart Arc Advanced Electric Wire Arc Thermal Spray System. The leading-edge gun head is integrated to a Robot - Arm and a Workpiece Manipulator (Turntable). The system produces highly reliable electric wire arc spray coatings for wear protection, restoration of component dimension and corrosion protection



FANUC R-1000iA/80F



5.2 Vertical lathe with a minimum capacity of
 Swing Ø: 1800mm
 Cutting Ø: 1800mm
 height: 1600mm



□VJNR/L

外径-端面切削

□VJNR/L



型号 Type	刀片 Insert	规格 Size(mm)	刀垫 Shim	中心销 Pin	压板 Clamp	压头螺钉 Clamp Screw	扳手 Wrench
MVJNR/L1616H16	VNCD1604C10	16 16 100 42 16 20	MV 1603	MCT513	HL 2414	MCS625	L2.0, L3.0
MVJNR/L2020K16		20 20 125 42 20 25					
MVJNR/L2525M16		25 25 150 42 25 32					
MVJNR/L3232P16		32 32 170 42 32 40					

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5.3 Sand blasting equipment, applied for cleaning and anchoring pattern before metallization



5.4 Stand manufacturing cost

Workstation	Time
Welding	22h
Drill	04h
Estimated Costs	670.0 \$USD

6.0 Supporting Attachments / References

6.1 Inspection and Salvage of Front Wheel Spindles for Certain Off-Highway Trucks (OHT) {0679, 4006, 4201, 4205} (SEBF9309-11)

6.2 Fundamentals of Arc Spray for Reconditioning Components {0679, 1000, 7000} (SEBF9238-18)

7.0 Related Best Practices

None

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8.0 Acknowledgements

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